Lecture 31.03: K-Means Clustering — Intuition & Iterative Mechanics











📌 Topic Overview

Comprehensive algorithmic derivation of K-Means clustering. Explains centroid initialization, assignment step via Euclidean distance, update step via cluster means, convergence conditions, and Ma'am's blackboard walkthrough.

📋 Key Learning Objectives

- Step 1: Choosing K (number of clusters) and initializing centroids C_1, ..., C_K.
- Step 2 (Assignment Step): Mapping each point x_i to its nearest centroid via Euclidean distance.
- Step 3 (Update Step): Recomputing centroids as the center of mass (arithmetic mean) of assigned points.
- Convergence Criteria: Centroid displacement threshold epsilon, assignment stability, or max_iter.
- Blackboard trace: Iteration-by-iteration movement of centroids.

🔍 In-Depth Conceptual Breakdown

- **K-Means Ka Naam**: 'K' ka matlab clusters ki sankhya, aur 'Means' ka matlab cluster ke points ka average (centroid).

- **Step 1: Initialize K Centroids**: Sabse pehle hum K centroids ($C_1, C_2, \ldots, C_K$) ko data space mein randomly place karte hain.

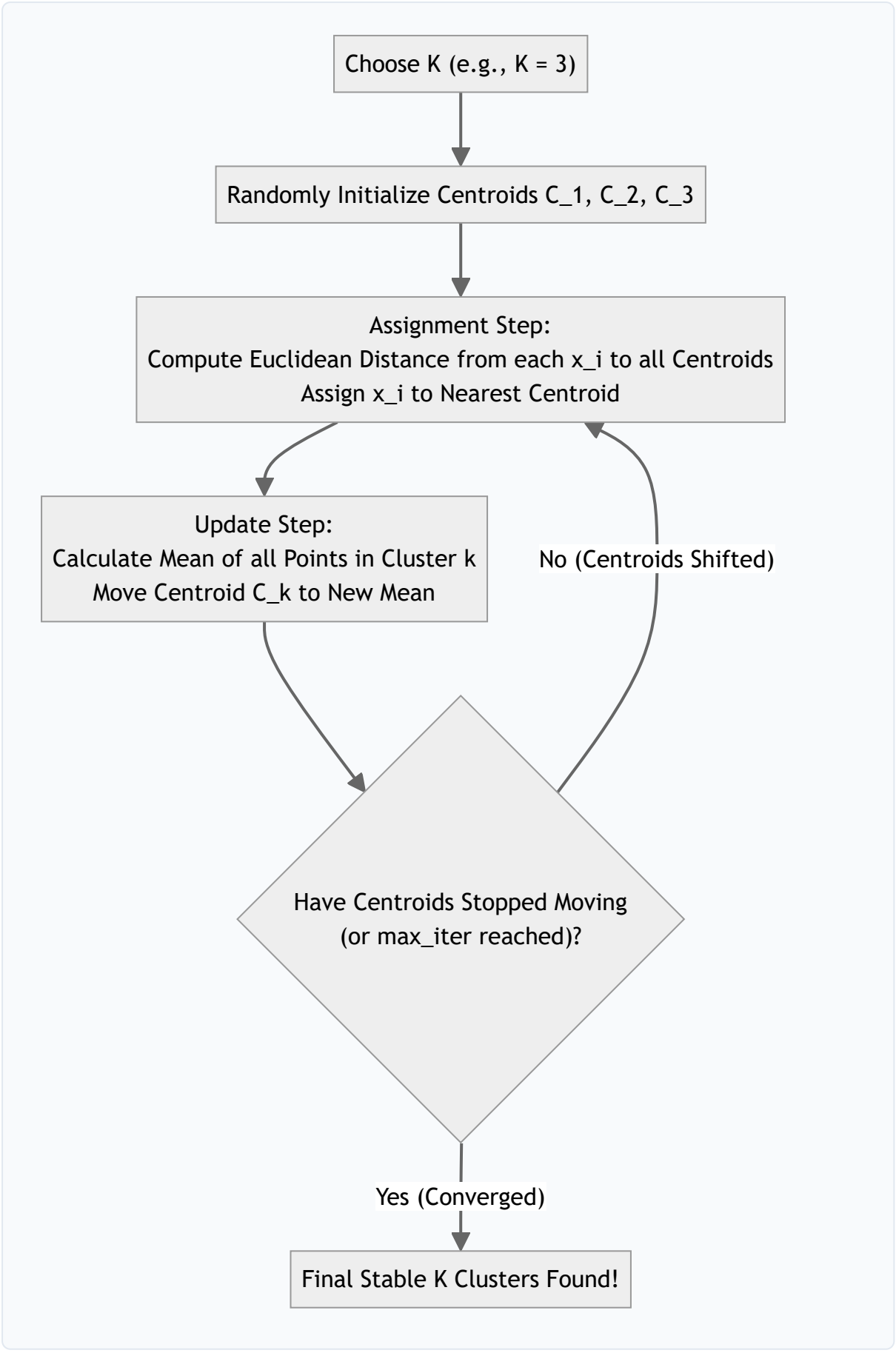
- **Step 2: Assignment Step (Sabse paas kaun?)**: Har data point x_i ka distance teeno centroids se nikalo: $d(x_i, C_1), d(x_i, C_2), d(x_i, C_3)$. Jis centroid se distance sabse kam hoga, point usi cluster ka part ban jata hai.
- **Step 3: Update Step (Naya Center)**: Har cluster ke andar jitne points assign hue, unka average (mean) nikalo: $C_k^{new} = \frac{1}{N_k} \sum x_i$. Centroid ab is naye mean par shift ho jata hai!
- **Step 4: Repeat Until Convergence**: Assignment aur Update steps ko tab tak repeat kiya jata hai jab tak:
    - Centroids ka movement band na ho jaye (shift ≈ 0),
    - Points ka cluster change hona band ho jaye, ya
    - Maximum iterations (`max_iter=300`) reach ho jayein.
- **Pure Theory Lecture**: Is lecture mein ma'am ne blackboard par step-by-step points assign karke dikhaye hain, koi code nahi hai.

📋 K-Means Algorithmic Lifecycle

Stage	Primary Operation	Mathematical Formula	Purpose
1. Initialization	Select K initial seeds	$\{C_1, C_2, \dots, C_K\} \subset \mathbb{R}^D$	Seed starting cluster locations
2. Assignment	Assign points to nearest centroid	$c_i = \arg\min_k \|\mathbf{x}_i - C_k\|^2$	Partition data into Voronoi cells
3. Update	Recompute cluster centers	$C_k = \frac{1}{\|S_k\|} \sum_{i \in S_k} \mathbf{x}_i$	Move centroids to cluster centers of mass
4. Convergence	Check displacement	$\|C_k^{(t+1)} - C_k^{(t)}\| < \epsilon$	Terminate optimization

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📊 Ma'am's K-Means Iterative Flowchart



🔢 Mathematical Formulation

Objective function (Inertia / Sum of Squared Errors):

$$J(\mathbf{C}) = \sum_{k=1}^{K} \sum_{\mathbf{x}_i \in S_k} \|\mathbf{x}_i - \mathbf{C}_k\|^2$$

Centroid update coordinate for feature dimension d:

$$C_{kd} = \frac{1}{|S_k|} \sum_{\mathbf{x}_i \in S_k} x_{id}$$

📌 Note on Code

This is a pure conceptual, mathematical, and blackboard theory lecture. In accordance with course guidelines, no synthetic code is appended.